Gennadii Filatov

filatovgwork@gmail.com gennadiifilatov.github.io GennadiiFilatov gennadii-filatov

Research interests: trustworthy generative modelling; synthetic tabular data; probabilistic machine learning; uncertainty quantification

Education

ITMO University, Saint Petersburg, Russia Sep 2025 – Present
M.Sc. in Applied Mathematics and Computer Science, Big Data and Machine Learning GPA: 4.00/4.00

- Thesis research: calibration of synthetic tabular data for trustworthy ML model selection; supervisor: Dr. Irina Deeva; advisor: Prof. Alexey Zaytsev.

Alexandru Ioan Cuza University of Iaşi (UAIC), Romania Sep 2026 – Present
Erasmus+ *Academic Exchange, Faculty of Computer Science* Selected participant

Peter the Great St. Petersburg Polytechnic University, Saint Petersburg, Russia Sep 2021 – Jun 2025
B.Sc. in Physics, Nuclear and Particle Physics GPA: 4.92/5.00

- Thesis: precision measurement of cathode-wire positions in the high-pressure proportional-chamber system for the PRES experiment; advisor: Dr. Polina Kravchenko.

Research Experience

ITMO University and HSE University Oct 2025 – Present
Machine Learning Researcher (supervisor: Dr. Irina Deeva) Saint Petersburg, Russia

- Developed two loss-based calibration methods: a constrained quadratic-program alignment of synthetic and real loss vectors, and a Bayesian Personalized Ranking (BPR)-style pairwise calibration that learn importance weights over synthetic validation samples so that weighted synthetic losses reproduce real held-out performance for model ranking.
- Evaluated across 10 benchmark datasets (5 classification, 5 regression), 122 model architectures, and 5 generative paradigms (CTGAN, TVAE, Gaussian Copula, TabDDPM, TabPFN-based); mean Spearman rank correlation with the true ranking improved from 0.37 to 0.73 (classification) and from 0.53 to 0.80 (regression). The work resulted in an AISTATS 2026 workshop publication.

SMILES-2026 International Summer School on Machine Learning Jul 2026
Selected Research Participant, Team Project Suzhou, China

- Selected through a competitive international process from 2,500+ applicants; 104 participants attended the in-person programme.
- Contributed to *Block Categorical Flow Maps*, a few-step semi-autoregressive text-generation method combining block-autoregressive modelling, categorical flow maps, and KV caching.
- Built experimental infrastructure and led a grid study over the number of function evaluations and block size; the team reduced generation perplexity from 166 to 49 in 16 sampling steps.

High Energy Physics Department, SPbPU Jul 2023 – Aug 2025
Senior Laboratory Assistant (supervisor: Dr. Polina Kravchenko) Saint Petersburg, Russia

- Developed image-processing methods for multi-wire high-pressure proportional chambers in the PRES experiment and analysed systematic measurement errors.
- Improved precision of cathode-wire position reconstruction by 30%, supporting proton charge-radius measurements.

Laboratory of Diagnostics of High-Temperature Plasma, SPbPU Feb – Jun 2023
Machine Learning Research Intern (supervisor: Dr. Alexander Yashin) Saint Petersburg, Russia

- Developed a 1D CNN in TensorFlow/Keras for automated detection of filaments in tokamak diagnostic signals, replacing manual signal-analysis steps.

Publications

G. Filatov, I. Deeva. *"When Synthetic Data Is Enough: Calibration for Tabular Model Ranking."* Workshop on Towards Trustworthy Predictions: Theory and Applications of Calibration for Modern AI, **AISTATS 2026**. [OpenReview]

Selected Presentations

- **Poster:** *When Synthetic Data Is Enough: Calibration for Tabular Model Ranking.* AISTATS 2026 Workshop, Morocco.
- **Poster:** *Calibration of synthetic tabular data for correct model selection.* XV Congress of Young Scientists, ITMO University, 2026.
- **Poster:** *Precision measurement of cathode-wire positions in a proportional high-pressure chamber system.* ICPS 2024, Georgia.

Honors & Awards

- **National Academic Scholarship for Science and Education, 2026–2027** — competitive merit-based scholarship in science and education, awarded by ITMO University.
- **Erasmus+ Mobility Grant, 2026–2027** — competitive EU-funded academic exchange at Alexandru Ioan Cuza University of Iaşi, Romania, awarded through a selective institutional process.
- **Enhanced State Academic Scholarship, 2022–2025** — awarded to students in the top 5% at SPbPU.
- **"Student of the Year," SPbPU, 2024** — university-wide distinction for academic, research, and extracurricular achievement.
- **IAPS Worldwide Grant, 2024** — competitive travel grant supporting participation in the International Conference of Physics Students.

Technical Skills

Programming: Python, C++, Bash, LaTeX, MATLAB
ML / statistics: PyTorch, scikit-learn, TensorFlow/Keras, NumPy, SciPy, pandas
Research tools: Git, Docker, Jupyter, Linux, VS Code
Languages: Russian (native), English (professional working proficiency)

Leadership, Teaching & Academic Service

AI Challenge Sep – Nov 2026
Solution Evaluator
Evaluator for Russia and the CIS's largest international AI competition for participants aged 25 and under, organized by a national industry consortium for AI. Assessed participant solutions against competition criteria across the main and final stages, reviewed code reproducibility and originality, and took part in live code-review sessions with participants.

Summer Scientific School "A5 School" Aug 2022 – Jul 2024
Co-Founder & Head of Education
Founded a science summer school; recruited and trained tutors, designed physics and mathematics curricula, and delivered lectures. More than 115 school students completed the program.

International Association of Physics Students (IAPS) Local Committee Saint Petersburg Apr 2023 – Apr 2025
President
Led a student outreach team; secured three grants totaling EUR 17,000 and organized 113+ educational events reaching 1,650+ students.